

Chapter 3

Electric Currents and Resistance



Electric Current $\Rightarrow$ total charge flowing through area per unit time

$$I = \frac{Q}{t}, \quad Q = ne, \quad e = 1.6 \times 10^{-19}$$

$$I = \frac{dQ}{dt} \text{ (instantaneous current)}$$

$$E = qV$$

Voltage, Current, Resistance

$$- V = IR$$

- Resistance $\Rightarrow$ a ratio of potential difference across an electrical component to the current flow through it

Resistivity $\Rightarrow$ resistance of a unit cross-sectional area per unit length of a material

$$- \rho = \frac{RA}{l} \quad \begin{array}{l} A = \text{area of cross section} \\ l = \text{length of wire} \end{array}$$

(Ωm)

$$- \rho_T = \rho_0 [1 + \alpha(T - T_0)], \quad \alpha = \text{temperature coefficient constant}$$

Power $\Rightarrow$ the energy liberated per unit time

$$\begin{aligned} - P &= \frac{W}{t} \\ - P &= VI \\ - P &= I^2 R \\ - P &= \frac{V^2}{R} \end{aligned}$$

Alternating current

- Average Power

$$\cdot \bar{P} = \frac{1}{2} I_0^2 R$$

$$\cdot \bar{P} = \frac{1}{2} \frac{V_0^2}{R}$$

$$\cdot \bar{P} = \frac{1}{2} I_0 V_0$$

$$- V_{rms} = 0.707 V_0$$

$$- I_{rms} = 0.707 I_0$$

$$- \text{Power (A.C.)} = V_{rms} \times I_{rms}$$

$$- V_{rms} = I_{rms} \times R, \quad R \text{ is constant}$$

- V_0 is peak voltage, V_{rms} is average

- I_0 is peak current, I_{rms} is average

Lesson outlines:

3-1 Electric battery

3.2 Electric current

3.3 Ohm's Law; Resistance and resistor

3.4 Resistivity

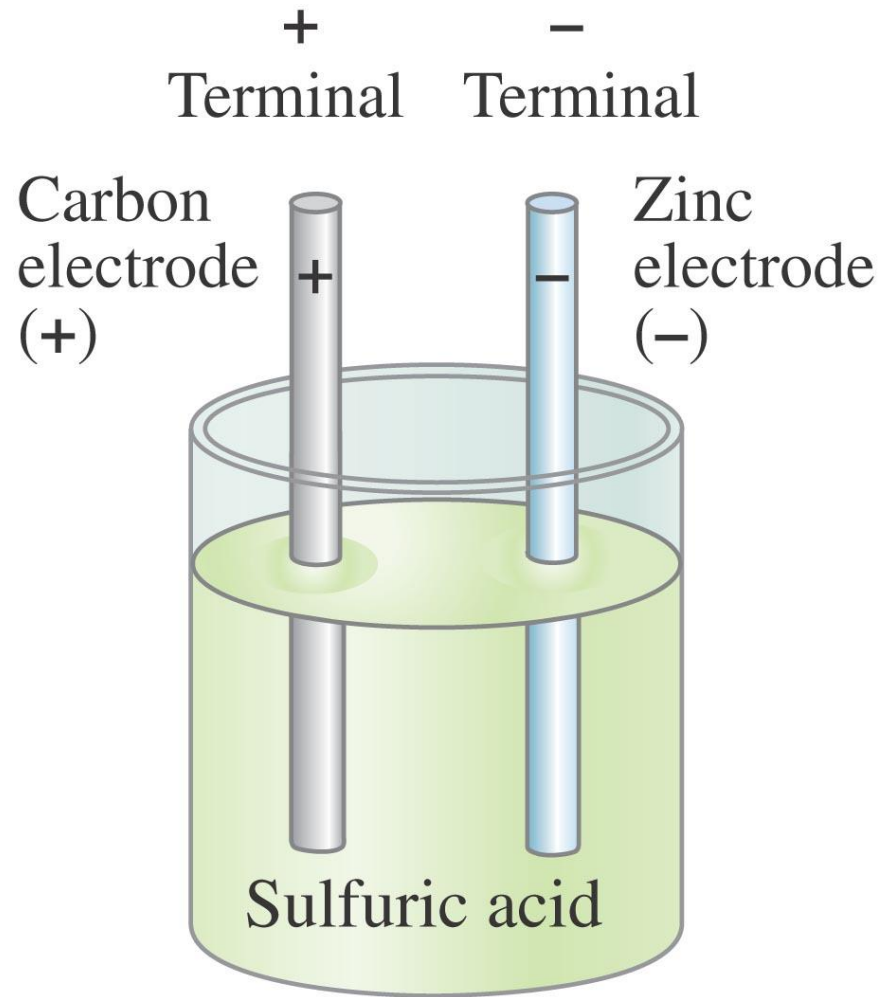
3.5 Electric power; power in household

3.6 Alternating Current

3-1 The Electric Battery

Volta discovered that electricity could be created if dissimilar metals were connected by a conductive solution called an electrolyte.

This is a simple electric cell.



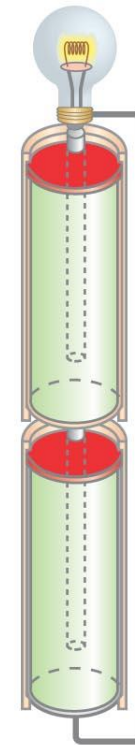
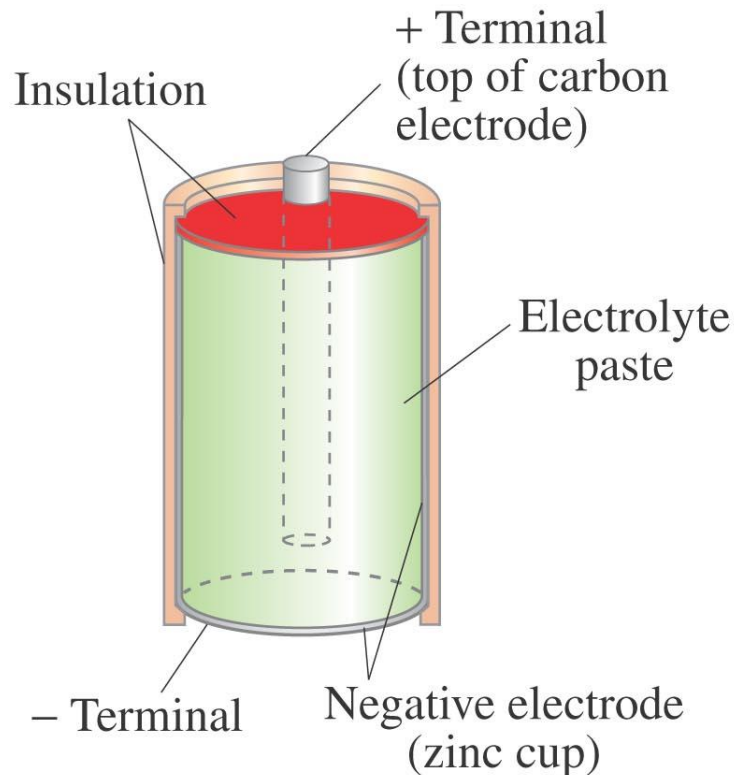
3-1 The Electric Battery

A battery transforms chemical energy into electrical energy.

Chemical reactions within the cell create a potential difference between the terminals by slowly dissolving them. This potential difference can be maintained even if a current is kept flowing, until one or the other terminal is completely dissolved.

3-1 The Electric Battery

Several cells connected together make a battery, although now we refer to a single cell as a battery as well.



3-2 Electric Current

Electric current, I

- is defined as the total (nett) charge, Q flowing through the area per unit time, t .

Mathematically,

$$I = \frac{Q}{t}$$

→ average current

OR

$$I = \frac{dQ}{dt}$$

→ instantaneous current

3-2 Electric Current

- It is a **base and scalar** quantities.
- The **S.I. unit** of the electric current is the **ampere (A)**.
- **1 ampere** of current is defined as **one coulomb of charge passing through the surface area in one second**.

OR

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{1 \text{ second}} = 1 \text{ C s}^{-1}$$

Note:

If the **charge move around a circuit in the same direction at all times**, the current is called **direct current (DC)**, which is **produced by the battery**.

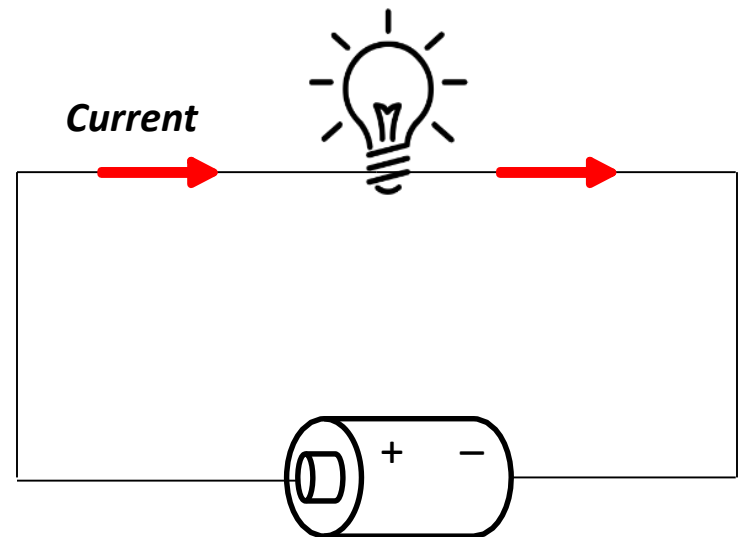
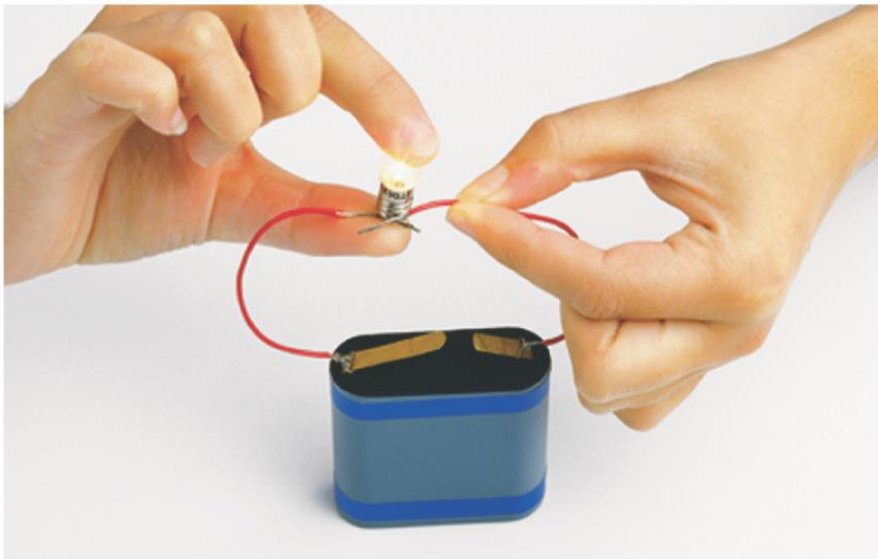
3-2 Electric Current

Complete circuit

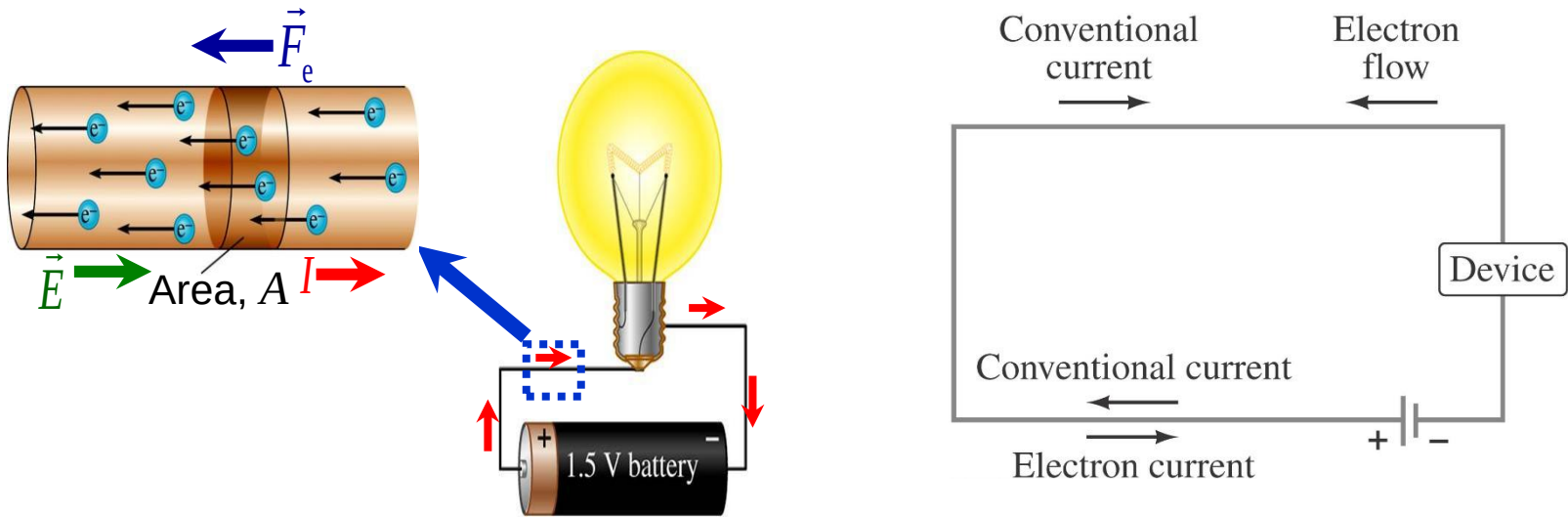
- is one where current can flow all the way around.

Open circuit

- is there a break in the circuit, or we it say cut wire / no current flows.



3-2 Electric Current



○ From the Figure

- Direction of electric field or electric current (conventional current) :
Positive to negative terminal
- Direction of electron flows :
Negative to positive terminal
- The **electron accelerates** because of the **electric force** acted on it.

3-2 Electric Current

Example 1: Current is flow of charge.

A steady current of 2.5 A exists in a wire for 4.0 min. (a) How much total charge passed by a given point in the circuit during those 4.0 min? (b) How many electrons would this be?

$$I = 2.5 \text{ A}, t = 4 \text{ min} = 4 \times 60 = 240 \text{ s}$$

$$(a) I = \frac{Q}{t}$$

$$\begin{aligned} Q &= It \\ &= 2.5(240) \\ &= 600 \text{ C} \end{aligned}$$

$$(b) \text{ charge of } 1 \text{ e}^- = 1.6 \times 10^{-19} \text{ C}$$

$$\begin{aligned} \text{num of e}^- &= \frac{600}{1.6 \times 10^{-19}} \\ &= 3.75 \times 10^{21} \text{ electrons} \end{aligned}$$

3-3 Ohm's Law: Resistance and Resistors

Ohm's law

- States that **the potential difference across a metallic conductor is proportional to the current flowing through it if its temperature is constant.**

Mathematically,

$$V \propto I \quad \text{where} \quad T = \text{constant}$$

Then

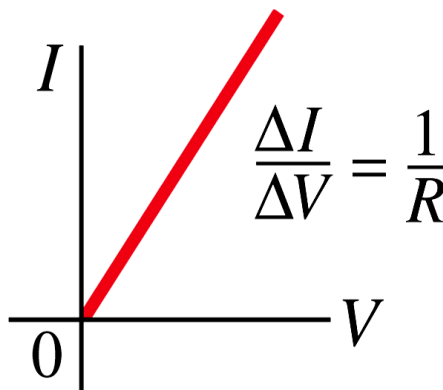
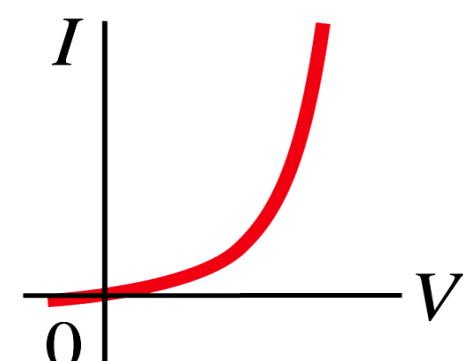
$$V = IR$$

where

R : resistance of a conductor

3-3 Ohm's Law: Resistance and Resistors

- Materials that obey Ohm's Law known as **Ohmic conductor**. (Eg. Pure metal).
 - The behavior of a piece of constantan wire is close to that of an Ohmic Conductor.
 - The graph of **current I** against **V** , the potential difference across the ohmic conductor, is a straight line through the origin.
- In many conductors, the resistance is independent of the voltage; this
- Materials that do **not follow Ohm's law** are called **non-ohmic**.

Ohmic conductor	Non-Ohmic conductor
 I $\frac{\Delta I}{\Delta V} = \frac{1}{R}$ 0 V	 I 0 V

3-3 Ohm's Law: Resistance and Resistors

Resistance, R

- is defined as **a ratio of the potential difference across an electrical component to the current passing through it.**

Mathematically,

$$R = \frac{V}{I}$$

where V : potential difference (voltage)
 I : current

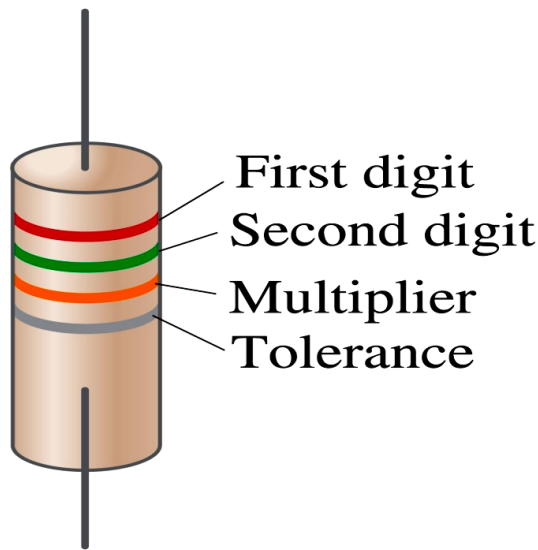
- It is a **measure of the component's opposition to the flow of the electric charge.**
- It is a **scalar** quantity and its unit is **ohm** (Ω) or **$V A^{-1}$**

Note:

If the **temperature** of the metallic conductor is **constant** hence its **resistance** also **constant**.

3-3 Ohm's Law: Resistance and Resistors

- Standard resistors are manufactured for use in electric circuits; they are color-coded to indicate their value and precision.



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Resistor Color Code			
Color	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
No color			20%

3-3 Ohm's Law: Resistance and Resistors

Example 2: Flashlight bulb resistance.

A small flashlight bulb draws 300 mA from its 1.5-V battery. (a) What is the resistance of the bulb? (b) If the battery becomes weak and the voltage drops to 1.2 V, how would the current change?

(a) $V = IR$

$$1.5 = 300 \times 10^{-3} (R)$$

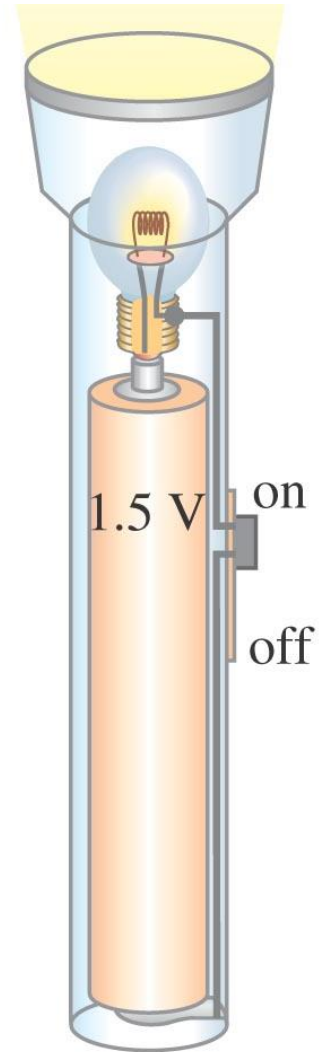
$$R = 5 \Omega$$

(b) $V = IR$

$$1.2 = I(5)$$

$$I = 0.24 \text{ A}$$

$\therefore$ the current decrease



3-4 Resistivity

Resistivity, ρ

- is defined as **the resistance of a unit cross-sectional area per unit length of the material.**

Mathematically,

$$\rho = \frac{RA}{l}$$

where

l : length of the material
 A : cross - sectional area

- It is **a scalar** quantity and its unit is **ohm meter ($\Omega \text{ m}$)**
- It is a **measure of a material's ability to oppose the flow of an electric current.**
- Resistivity depends on the **type of the material** and on the **temperature.**
- A good electric **conductors** have a very **low resistivities** and good **insulators** have very **high resistivities.**

3-4 Resistivity

- From the eq., the **resistance** of a conductor depends on the **length and cross-sectional area**.
- Table 1 shows the resistivity for various materials at 20 °C.

Material	Resistivity, ρ ($\Omega \cdot \text{m}$)	Temperature Coefficient, α ($^{\circ}\text{C}^{-1}$)
<i>Conductors</i>		
Silver	1.59×10^{-8}	0.0061
Copper	1.68×10^{-8}	0.0068
Gold	2.44×10^{-8}	0.0034
Aluminum	2.65×10^{-8}	0.00429
Tungsten	5.60×10^{-8}	0.0045
Iron	9.71×10^{-8}	0.00651
Platinum	10.60×10^{-8}	0.003927
Mercury	98.00×10^{-8}	0.0009
Nichrome (Ni, Fe, Cr alloy)	100.00×10^{-8}	0.0004
<i>Semiconductors[†]</i>		
Carbon (graphite)	$(3 - 60) \times 10^{-5}$	-0.0005
Germanium	$(1 - 500) \times 10^{-3}$	-0.05
Silicon	0.1 – 60	-0.07
<i>Insulators</i>		
Glass	$10^9 - 10^{12}$	
Hard rubber	$10^{13} - 10^{15}$	

[†] Values depend strongly on the presence of even slight amounts of impurities.

3-4 Resistivity

Example 3

The resistivity of a silver wire with a radius of $5.04 \times 10^{-4} \text{ m}$ is $1.59 \times 10^{-8} \Omega \cdot \text{m}$. If the length of the wire is 3.0 m , what is the resistance of the wire? (Ans: 0.0598Ω)

Solution

$$\rho = 1.59 \times 10^{-8} \Omega \cdot \text{m} \quad \ell = 3.0 \text{ m} \quad r = 5.04 \times 10^{-4} \text{ m}$$

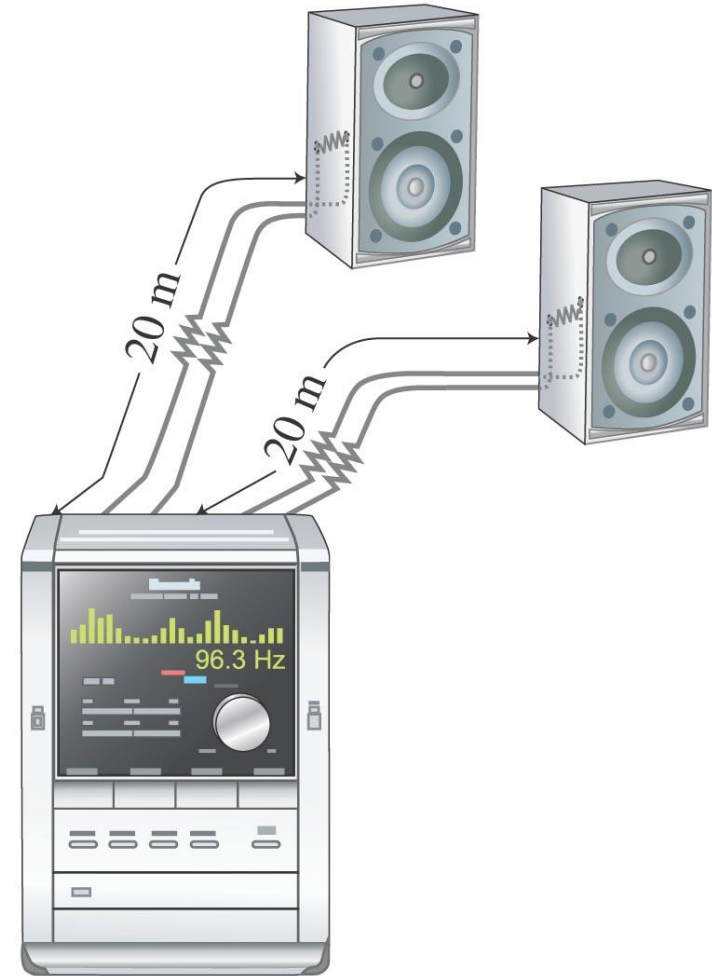
$$\begin{aligned} R &= \rho \frac{\ell}{A} = (1.59 \times 10^{-8} \Omega \cdot \text{m}) \left(\frac{3 \text{ m}}{7.98 \times 10^{-7} \text{ m}^2} \right) \\ &= 5.98 \times 10^{-2} \Omega \end{aligned}$$

3-4 Resistivity

Example 4: Speaker wires.

$\rho = 1.68 \times 10^{-8}$

Suppose you want to connect your stereo to remote speakers. (a) If each wire must be 20 m long, what diameter copper wire should you use to keep the resistance less than $0.10 \, \Omega$ per wire? (b) If the current to each speaker is 4.0 A, what is the potential difference, or voltage drop, across each wire?



$$(a) \rho = \frac{RA}{l}$$

$$1.68 \times 10^{-8} = \frac{0.1 (\pi r^2)}{20}$$

$$r = 1.034 \times 10^{-3} \text{ m}$$

$$d = 2r = 2.068 \times 10^{-3} \text{ m}$$
$$= 2.07 \text{ mm}$$

$$(b) I = 4 \text{ A}$$

$$V = IR$$

$$= 4(0.1)$$

$$= 0.4 \text{ V}$$

3-4 Resistivity

For any given material, the resistivity increases with temperature:

$$\rho_T = \rho_0[1 + \alpha(T - T_0)].$$

Semiconductors are complex materials, and may have resistivities that decrease with temperature.

3-4 Resistivity

Example 5: Resistance thermometer.

The variation in electrical resistance with temperature can be used to make precise temperature measurements. Platinum is commonly used since it is relatively free from corrosive effects and has a high melting point. Suppose at 20.0°C the resistance of a platinum resistance thermometer is 164.2 Ω . When placed in a particular solution, the resistance is 187.4 Ω . What is the temperature of this solution?

$$\alpha = 0.003927$$

$$S_T = S_0 [1 + \alpha(T - T_0)]$$

$$187.4 = 164.2 [1 + 0.003927(T - 20)]$$

$$1.1413 = 1 + 0.003927T - 0.07854$$

$$T = 55.98^\circ\text{C}$$

3-5 Electric Power

Power, P

- is defined as **the energy liberated per unit time in the electrical device.**
- The electrical power P supplied to the electrical device is given by

$$P = \frac{W}{t} = \frac{VIt}{t}$$

$$P = VI$$

- When the electric current flows through wire or passive resistor, hence the potential difference across it is

$$V = IR$$

then the electrical power can be written as

$$P = I^2 R$$

OR

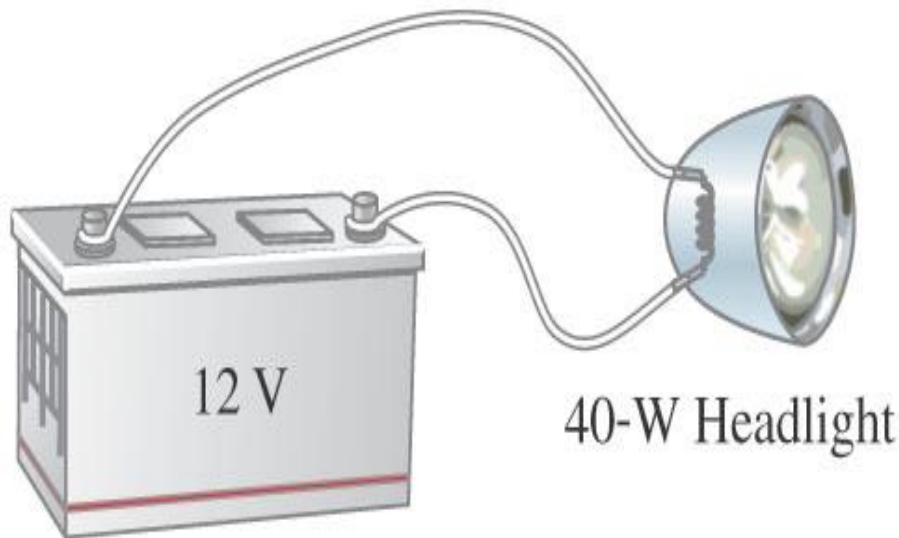
$$P = \frac{V^2}{R}$$

- It is a **scalar** quantity and its unit is **watts (W)**.

3-5 Electric Power

Example 6: Headlights.

Calculate the resistance of a 40-W automobile headlight designed for 12 V.



$$V = 12$$

$$P = \frac{V^2}{R}$$

$$40 = \frac{12^2}{R}$$

$$R = 3.6 \Omega$$

3-5 Electric Power

What you pay for on your electric bill is not power, but energy – the power consumption multiplied by the time.

We have been measuring energy in joules, but the electric company measures it in kilowatt-hours, kWh:

$$1 \text{ kWh} = (1000 \text{ W})(3600 \text{ s}) = 3.60 \times 10^6 \text{ J}.$$

3-5 Electric Power

Example 7: Electric heater.

An electric heater draws a steady 15.0 A on a 120-V line. How much power does it require and how much does it cost per month (30 days) if it operates 3.0 h per day and the electric company charges 9.2 cents per kWh?

$$30 \times 3 = 90 \text{ hours}$$

$$P = VI$$

$$= 120(15)$$

$$= 1800 \text{ W}$$

$$= 1.8 \text{ kW}$$

$$1.8 \times 90 \times 9.2 = 1490.4 \text{ cents}$$

$$= \$14.90$$

3-5 Electric Power

Example 8: Lightning bolt.

Lightning is a spectacular example of electric current in a natural phenomenon. There is much variability to lightning bolts, but a typical event can transfer 10^9 J of energy across a potential difference of perhaps 5×10^7 V during a time interval of about 0.2 s. Use this information to estimate (a) the total amount of charge transferred between cloud and ground, (b) the current in the lightning bolt, and (c) the average power delivered over the 0.2 s.

$$E = 1 \times 10^9 \text{ J}$$

$$V = 5 \times 10^7 \text{ V}$$

$$t = 0.2 \text{ s}$$

$$E = qV$$

$$1 \times 10^9 = q(5 \times 10^7)$$

$$q = 20 \text{ C}$$

$$Q = It$$

$$20 = I(0.2)$$

$$I = 100 \text{ A}$$

$$P = VI$$

$$= 5 \times 10^7 (100)$$

$$= 5 \times 10^9 \text{ W}$$

3-5.1 Power in Household Circuits

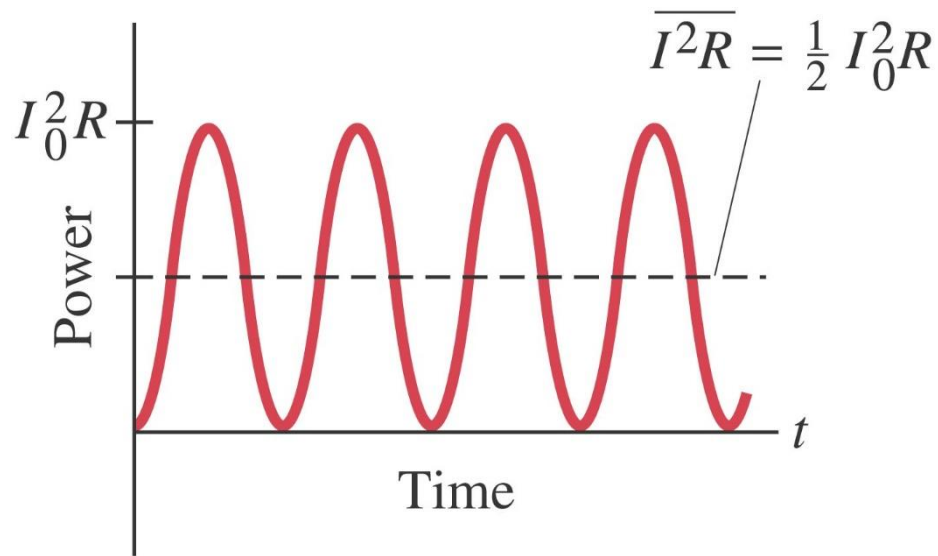
The wires used in homes to carry electricity have very low resistance. However, if the current is high enough, the power will increase and the wires can become hot enough to start a fire.

To avoid this, we use fuses or circuit breakers, which disconnect when the current goes above a predetermined value.

3-6 Alternating Current

Multiplying the current and the voltage gives the power:

$$P = I^2 R = I_0^2 R \sin^2 \omega t.$$



Usually we are interested in the average power:

$$\overline{P} = \frac{1}{2} I_0^2 R$$

$$\overline{P} = \frac{1}{2} \frac{V_0^2}{R}.$$

3-6 Alternating Current

The current and voltage both have average values of zero, so we square them, take the average, then take the square root, yielding the root-mean-square (rms) value:

$$I_{\text{rms}} = \sqrt{\overline{I^2}} = \frac{I_0}{\sqrt{2}} = 0.707 I_0,$$

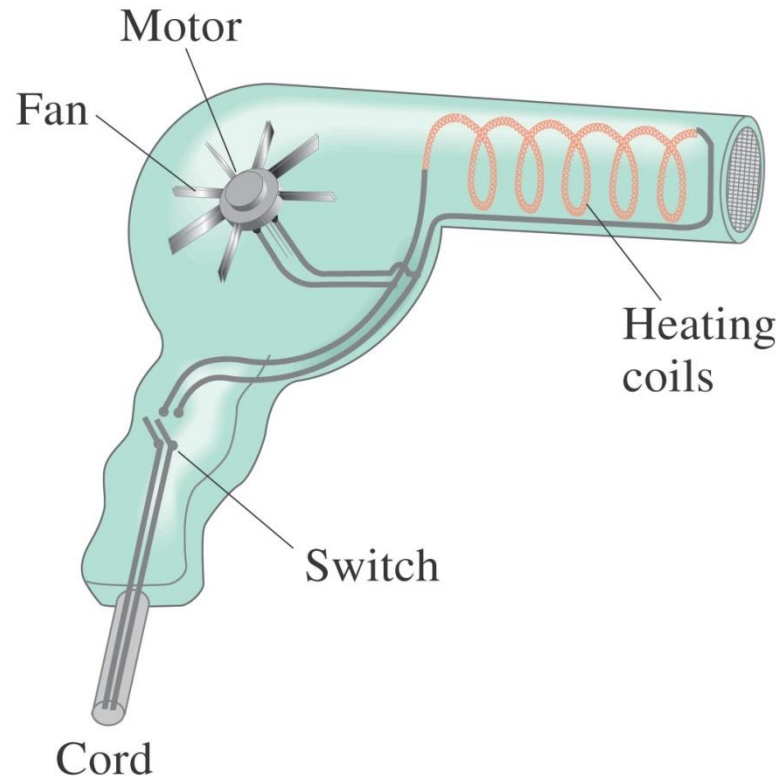
$$V_{\text{rms}} = \sqrt{\overline{V^2}} = \frac{V_0}{\sqrt{2}} = 0.707 V_0.$$

$$\text{Power (AC)} = V_{\text{rms}} \times I_{\text{rms}}$$

3-6 Alternating Current

Example 9: Hair dryer.

- (a) Calculate the resistance and the peak current in a 1000-W hair dryer connected to a 120-V line.
- (b) What happens ^{to power} if it is connected to a 240-V line in Britain?



$$(9) \quad P = 1000 \text{ W}$$

$$V_{\text{rms}} = 120$$

$$P = VI$$

$$1000 = 120(I)$$

$$I_{\text{rms}} = 8.33 \text{ A}$$

$$I_0 = I_{\text{rms}} \times \sqrt{2}$$

$$= 11.78 \text{ A}$$

(maximum current)

$$V_{\text{rms}} = I_{\text{rms}} R$$

$$120 = 8.33(R)$$

$$R = 14.4 \, \Omega$$

$$(6) \quad P = \frac{V_{\text{rms}}^2}{R}$$
$$= \frac{240^2}{14.4}$$
$$= 4000 \text{ W}$$

Summary of Chapter 3

- A battery is a source of constant potential difference.
- Electric current is the rate of flow of electric charge.
- Conventional current is in the direction that positive charge would flow.
- Resistance is the ratio of voltage to current:

$$I = \frac{V}{R},$$

$$V = IR.$$

Summary of Chapter 3

- Ohmic materials have constant resistance, independent of voltage.
- Resistance is determined by shape and material:

$$R = \rho \frac{\ell}{A}.$$

- ρ is the resistivity.

Summary of Chapter 3

- Power in an electric circuit:

$$P = IV.$$

- Direct current is constant.
- Alternating current varies sinusoidally:

$$I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = I_0 \sin \omega t.$$

Summary of Chapter 3

- The average (rms) current and voltage:

$$I_{\text{rms}} = \sqrt{\overline{I^2}} = \frac{I_0}{\sqrt{2}} = 0.707 I_0,$$

$$V_{\text{rms}} = \sqrt{\overline{V^2}} = \frac{V_0}{\sqrt{2}} = 0.707 V_0.$$